

Question	Answer	Marks
3(a)	$W = Fd$	B1
	$P = Fd / t = Fv$ or $P = Fvt / t = Fv$	B1
3(b)(i)	$(\Delta)E_{(P)} = mg(\Delta)h$	C1
	increase of gravitational potential energy of car in 1.0 s $= 1500 \times 9.81 \times 30 \times \sin 6.0 = 46\,000 \text{ J}$	A1

Question	Answer	Marks
3(b)(ii)	(Power to overcome total resistive forces) $= 1600 \times 30$ $= 48\,000\text{ W}$	C1
	power $= 48\,000 + 46\,000$ $= 9.4 \times 10^4\text{ W}$	A1
3(c)(i)	Air resistance is the same, as the speed is the same	B1
3(c)(ii)	Mass / weight has increased so (power will) increase	B1

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